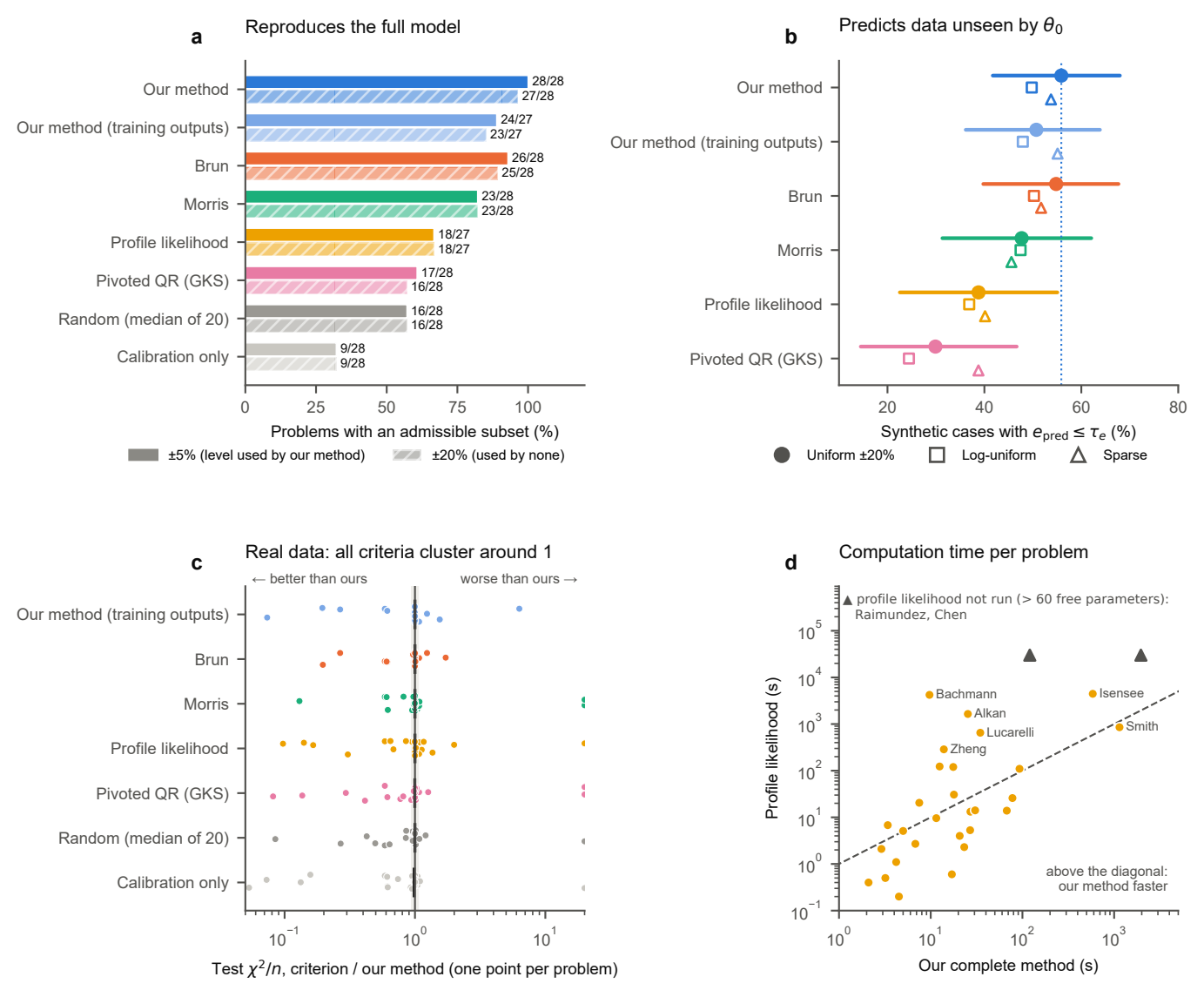

a
Reproduces the full model
Our method
Our method (training outputs)
Brun
Morris
Profile likelihood
Pivoted QR (GKS)
Random (median of 20)
Calibration only
28/28
27/28
24/27
23/27
26/28
25/28
23/28
23/28
18/27
18/27
17/28
16/28
16/28
16/28
9/28
9/28
0
25
50
75
100
Problems with an admissible subset (%)
±5% (level used by our method)
±20% (used by none)
b
Predicts data unseen by θ_0
Our method
Our method (training outputs)
Brun
Morris
Profile likelihood
Pivoted QR (GKS)
20
40
60
80
Synthetic cases with $e_{\text{pred}} \le \tau_e$ (%)
Uniform ±20%
Log-uniform
Sparse
c
Real data: all criteria cluster around 1
← better than ours
worse than ours →
Our method (training outputs)
Brun
Morris
Profile likelihood
Pivoted QR (GKS)
Random (median of 20)
Calibration only
10^{-1}
10^{0}
10^{1}
Test χ^2/n, criterion / our method (one point per problem)
d
Computation time per problem
▲ profile likelihood not run (> 60 free parameters):
Raimundez, Chen
Bachmann
Isensee
Alkan
Lucarelli
Smith
Zheng
above the diagonal:
our method faster
Profile likelihood (s)
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
Our complete method (s)